DIRECTORATE OF GOVERNMENT EXAMINATIONS, CHENNAI -6

HSE SECOND YEAR EXAMINATIONS, MARCH 2025

CHEMISTRY ANSWER KEY - ENGLISH MEDIUM

Maximum Marks : 70

GENERAL INSTRUCTIONS

1. Answer written only in BLACK or BLUE should be evaluated.
2. Choose the most suitable answer in Part I from the given alternatives and write the option code and the corresponding answer.

PART - I

Answer all the questions **15×1=15**

A TYPE			B TYPE			MARK
Q. No	Option code	Answer	Q. No	Option code	Answer	
1	c	3.0 mol min^{-1}	1	c	α –helical backbone	1
2	a	3	2	a	N- methylaniline	1
3	c	1, 1, 2, 2, - tetra fluoro ethane	3	a	6.022×10^{22}	1
4	a	N- methylaniline	4	a	8 and 4	1
5	c	accepts OH^- from water, releasing proton	5	c	3.0 mol min^{-1}	1
6	a	(iii)>(ii) > (i) > (iv)	6	b	+2	1
7	c	Both assertion and reason are false	7	a	4 - nitro phenol	1
8	a	8 and 4	8	c	Both assertion and reason are false	1
9	c	α –helical backbone	9	c	accepts OH^- from water, releasing proton	1
10	c	2.71×10^{-14}	10	c	acetyl chloride	1
11	b	+2	11	a	Displacement with zinc	1
12	a	6.022×10^{22}	12	c	2.71×10^{-14}	1
13	a	4 - nitro phenol	13	a	3	1
14	c	acetyl chloride	14	c	1, 1, 2, 2, - tetra fluoro ethane	1
15	a	Displacement with zinc	15	a	(iii)>(ii) > (i) > (iv)	1

PART-II

Answer **Any Six** Questions and

Question No.**24** Compulsory **6×2=12**

Q. No.	Answers	Marks
16.	Sulphide ore → 1 Mark Galena (PbS) (or) Zinc blende (ZnS) (or) any other sulphide (name (or) formula) → 1 Mark	2
17.	$PCl_5 \rightarrow PCl_3 + Cl_2$ Only Explanation ---------> 1 Mark	2
18.	**Zr & Hf similar properties** Lanthanoid contraction (or) Similar atomic radii (or) Similar ionic radii (or) Similar size (or) very small change in radii (or) any correct explanation	2
19.	**Solubility product** Correct definition (or) Formula	2
20.	**Equivalent conductance** Correct definition (or) Formula	2
21.	Explanation (or) any one correct equation ----------> 2 Marks Without mentioning peptising agent ----------> 1 Mark	2
22.	**Gattermann Koch reaction** Correct equation ----------> 2 Marks Explanation only ----------> 1 Mark Not mentioning $AlCl_3$/CuCl ----------> 1½ Mark Not mentioning CO,HCl, $AlCl_3$/CuCl ----------> 1 Mark	2
23.	**Classification of Drugs** - Four types → 4×½	2
24.	X = C_6H_6 (or) Benzene Y = C_6H_5 - NH_2 (or) Aniline	1 1

PART-III

Answer **Any Six** Questions and
Question No.**33** is Compulsory **6×3=18**

Q. No.	Answer		Marks
25.	**Refining of Ni** 2 Equations with condition Explanation only (or) Equation without condition Mentioning Mond's process	 3 2 1	3
26.	**Bleaching action of Sulphur dioxide** $SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2[H]$ $X + 2[H] \longrightarrow XH_2$ coloured colourless (or) In presence of water, Sulphur dioxide bleaches coloured wool, silk,sponges and straw into colourless due to its reducing property. ------ **1 Mark** Bleached product (colourless) is reoxidised by atmospheric oxygen to its original colour ------ **1 Mark** Bleachig action of SO_2 is temporary ------- **1 Mark**		1½ 1½
27.	**Hydrate isomers** - Correct Explanation Example (any 2 isomers)		2 1
28.	**Distinguish tetrahedral and octahedral voids** - Any 3 differences		3
29.	**Limitations of Freundlich adsorption isotherm** - Any 2 limitations		2×1½
30.	A - $C_6H_5NH_2$ (or) Amino benzene (or) Aniline (or) Structure B - $C_6H_5N_2Cl$ (or) Benzene Diazonium Chloride (or) Structure C - C_6H_5CN (or) Benzene Nitrile (or)Phenyl Cyanide (or) Cyanobenzene (or) Structure		1 1 1
31.	**Condensation polymer** - Correct definition Any 2 Examples		2 ½+½
32.	**Functions of Lipids in living organism** - Any 3 functions		3×1

33	**Half life time – calculation** $t_{1/2} = \frac{0.6932}{k}$ (or) $k = 0.6932/t_{1/2}$ $= \frac{0.6932}{1.54 \times 10^{-3}}$ = 450 sec	1 1 ½ + ½

PART-IV

Answer All Questions **5×5=25**

Q. No.	Answer	Marks
34. (a)	**(i) Cryolite in the extraction of aluminium :** Lowering of melting point **Iodine in the refining of zirconium :** To form easily volatile compounds (or) Correct equation **ii) Properties of inter halogen compound :** Any 3 Properties	1 1 3
	(OR)	
34 (b)	**Identification of Borate radicals** i) Correct equation Unbalanced equation (or) without mentioning con H_2SO_4 → **1½ Mark** Explanation only ------> **1 Mark** Ethyl borate test mentioned (or) Green edged flame ------> **½ Mark** ii) Any 3 uses	2 3
35. (a)	**Preparation of potassium dichromate :** $4FeCr_2O_4 + 8\,Na_2CO_3 + 7O_2 \xrightarrow{900\text{-}1000^oC} 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$ → **2 Mark** without temperature → **1½ Mark** Unbalanced equation → **1½ Mark** $2Na_2CrO_4 . H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$ → **1½ Mark** Unbalanced equation → **1 Mark** $Na_2Cr_2O_7 . 2KCl \longrightarrow K_2Cr_2O_7 . 2NaCl$ → **1½ Mark** Unbalanced equation → **1 Mark** Explanation only → **3 Mark** (or) Mentioning ore and gravity separation → **1 Mark**	5
	(OR)	

35 (b)	**Postulates of werners theory:** Any 5 postulates only diagram ------ > **1 Mark**	5
36. (a)	**i) Metal excess defect** Explanation (Diagram not necessary) → **1 Mark** Any one example → **1 Mark** Diagram only → **½ Mark** **ii) Pseudo first order reaction** Correct definition → **2 Mark** Correct Example (Explanation or Equation) → **1 Mark**	5
	(OR)	
36 (b)	$[H^+]\ (or)[H_3O^+] = \sqrt{Ka\ X\ C}$ $= \sqrt{1.8\ X\ 10^{-5}\ X\ 0.1}$ $= 1.34 \times 10^{-3}$ M pH = -log $[H^+]$ (or) $-log[H_3O^+]$ pH = -log $[1.34 \times 10^{-3}]$ (or) = 3-log 1.34 = 2.87 **(or)** Any other relevant method ------ > **5 Marks**	1 ½ ½+½ 1 1 ½
37. (a)	Nernst equation : $xA + yB \rightleftharpoons lC + mD$ $\Delta G = \Delta G^\circ + RT \ln Q$ $Q = \frac{[C]^l[D]^m}{[A]^x[B]^y}$ (or) $\Delta G = \Delta G^\circ + RT \ln \frac{[C]^l[D]^m}{[A]^x[B]^y}$ $\Delta G = -nFE_{Cell}$; $\Delta G^\circ = -nFE^\circ_{Cell}$ (or) $-nFE_{Cell} = -nFE^\circ_{Cell} + RT \ln \frac{[C]^l[D]^m}{[A]^x[B]^y}$ $E_{Cell} = E^\circ_{Cell} - \frac{RT}{nF} \ln \frac{[C]^l[D]^m}{[A]^x[B]^y}$ (or) $E_{Cell} = E^\circ_{Cell} - \frac{2.303\,RT}{nF} \log \frac{[C]^l[D]^m}{[A]^x[B]^y}$ (or) $E_{Cell} = E^\circ_{Cell} - \frac{0.0591}{n} \log \frac{[C]^l[D]^m}{[A]^x[B]^y}$	1 1 1 1 1
	(OR)	

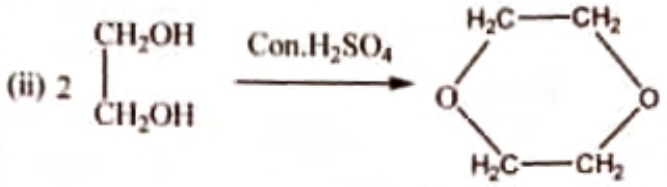
37 (b)

(i) Ethyl alcohol to Ethene

$$CH_3 - CH_2OH \xrightarrow[443K]{Con.H_2SO_4\ (or)\ Phosphoric\ acid\ (or)\ anhydrou\ ZnCl_2\ (or)\ Alumina} CH_2 = CH_2$$ — 2

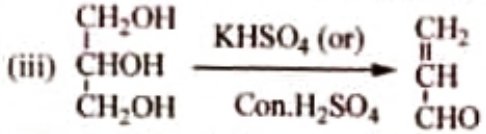
without dehydrating agent (or) without temp → **1½ Mark**

Explanation only → **1 Mark**

(ii) 

— 1½

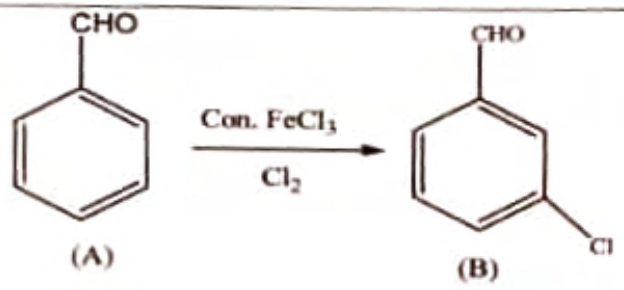
without dehydrating agent (or) Explanation only → **1 Mark**

(iii) 

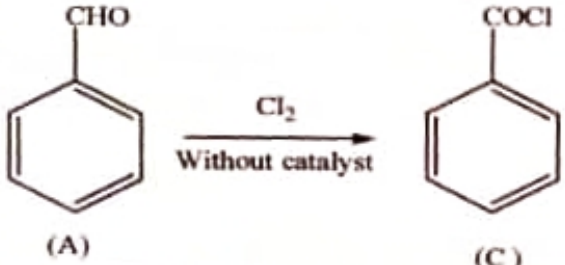
— 1½

without dehydrating agent (or) Explanation only → **1 Mark**

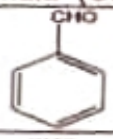
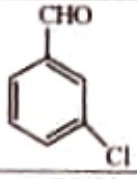
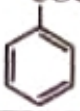
38 (a)



— 1



— 1

Compound	Formula (OR)	Name
A		Benzaldehyde(or) Phenylmethanal
B		m-Chloro benzaldehyde (or) 3-Chlorophenylmethanal
C		Benzoyl chloride

— 1 (A), 1 (B), 1 (C)

If A,B, C mentioned in the correct equation itself → **5 Marks**

(OR)

38 (b)	**(i) Nitro form and Aci form** Any two differences (or) If two forms structure mentioned → **1 Mark** **ii) Types of RNA** • Ribosomal RNA (or) (rRNA) • Messenger RNA (or) (mRNA) • Transfer RNA (or) (tRNA)	2 3